

ANALYTICAL REPORT

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Laboratory Job ID: 190-22815-1

Client Project/Site: City of Cadillac Semiannual PFAS

For:

City of Cadillac Utilities
1121 Plett Road
Cadillac, Michigan 49601

Attn: Cindy Tomaszewski

Sue Schafer

Authorized for release by:
5/5/2020 2:09:29 PM

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Results relate only to the items tested and the sample(s) as received by the laboratory.

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Sample Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-22815-1

Lab Sample ID	Client Sample ID	Matrix	Collected	Received	Asset ID
190-22815-1	Field Blank	Water	04/22/20 14:40	04/23/20 10:24	
190-22815-2	Effluent	Water	04/22/20 14:45	04/23/20 10:24	
190-22815-3	Effluent Duplic.	Water	04/22/20 14:50	04/23/20 10:24	

Case Narrative

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-22815-1

Job ID: 190-22815-1

Laboratory: Eurofins TestAmerica, Michigan

Narrative

Job Narrative 190-22815-1

Comments

No additional comments.

Receipt

The samples were received on 4/23/2020 10:23 AM; the samples arrived in good condition, properly preserved and, where required, on ice. The temperature of the cooler at receipt was 2.0° C.

LCMS

Method 537 (modified): The laboratory control sample (LCS) associated with preparation batch 200-154582 and analytical batch 200-154592 was outside acceptance criteria for Perfluorotridecanoic acid (PFTrIA). The batch matrix spike/matrix spike duplicate (MS/MSD) was within acceptance limits and may be used to evaluate matrix performance.

Method 537 (modified): M2-4:2 FTS Isotope Dilution Analyte (IDA) recovery is above the method recommended limit for the following sample: Effluent (190-22815-2). Quantitation by isotope dilution generally precludes any adverse effect on data quality due to elevated IDA recoveries.

Method 537 (modified): 13C2 PFTeDA Isotope Dilution Analyte (IDA) recovery associated with the following sample is below the method recommended limit: Effluent Duplic. (190-22815-3). Generally, data quality is not considered affected if the IDA signal-to-noise ratio is greater than 10:1, which is achieved for all IDA in the sample(s). All detection limits are below the lower calibration.

No additional analytical or quality issues were noted, other than those described above or in the Definitions/Glossary page.

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-22815-1

Client Sample ID: Field Blank

Lab Sample ID: 190-22815-1

Date Collected: 04/22/20 14:40

Matrix: Water

Date Received: 04/23/20 10:24

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4:2 FTS	<18		18	ng/L		04/30/20 11:33	04/30/20 16:33	1
6:2 FTS	<18		18	ng/L		04/30/20 11:33	04/30/20 16:33	1
8:2 FTS	<18		18	ng/L		04/30/20 11:33	04/30/20 16:33	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<18		18	ng/L		04/30/20 11:33	04/30/20 16:33	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<18		18	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluorobutanesulfonic acid (PFBS)	<1.8		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluorobutanoic acid (PFBA)	<1.8		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluorodecanesulfonic acid (PFDS)	<1.8		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluorodecanoic acid (PFDA)	<1.8		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluorododecanoic acid (PFDoA)	<1.8		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluoroheptanesulfonic Acid (PFHpS)	<1.8		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluoroheptanoic acid (PFHpA)	<1.8		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluorohexanesulfonic acid (PFHxS)	<1.8		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluorohexanoic acid (PFHxA)	<1.8		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluorononanesulfonic acid (PFNS)	<1.8		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluorononanoic acid (PFNA)	<1.8		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluorooctanesulfonamide (FOSA)	<9.0		9.0	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluorooctane sulfonate (PFOS)	<1.8		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluorooctanoic acid (PFOA)	<1.8		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluoropentanesulfonic acid (PFPeS)	<1.8		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluoropentanoic acid (PFPeA)	<1.8		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluorotetradecanoic acid (PFTeA)	<1.8		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluorotridecanoic acid (PFTrIA)	<1.8 *		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1
Perfluoroundecanoic acid (PFUnA)	<1.8		1.8	ng/L		04/30/20 11:33	04/30/20 16:33	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	67		25 - 150	04/30/20 11:33	04/30/20 16:33	1
13C4 PFBA	113		25 - 150	04/30/20 11:33	04/30/20 16:33	1
13C3 PFBS	110		50 - 150	04/30/20 11:33	04/30/20 16:33	1
13C2 PFDA	93		50 - 150	04/30/20 11:33	04/30/20 16:33	1
13C2 PFDoA	83		50 - 150	04/30/20 11:33	04/30/20 16:33	1
13C4 PFHpA	111		50 - 150	04/30/20 11:33	04/30/20 16:33	1
13C2 PFHxA	117		50 - 150	04/30/20 11:33	04/30/20 16:33	1
13C5 PFNA	99		50 - 150	04/30/20 11:33	04/30/20 16:33	1
13C4 PFOA	106		50 - 150	04/30/20 11:33	04/30/20 16:33	1
13C4 PFOS	108		50 - 150	04/30/20 11:33	04/30/20 16:33	1
13C5-PFPeA DNU	116		25 - 150	04/30/20 11:33	04/30/20 16:33	1
13C2 PFTeA	71		50 - 150	04/30/20 11:33	04/30/20 16:33	1
13C2 PFUnA	80		50 - 150	04/30/20 11:33	04/30/20 16:33	1
d5-NEtFOSAA	77		50 - 150	04/30/20 11:33	04/30/20 16:33	1
d3-NMeFOSAA	77		50 - 150	04/30/20 11:33	04/30/20 16:33	1
M2-4:2 FTS	125		25 - 150	04/30/20 11:33	04/30/20 16:33	1
M2-6:2 FTS	108		25 - 150	04/30/20 11:33	04/30/20 16:33	1
M2-8:2 FTS	105		25 - 150	04/30/20 11:33	04/30/20 16:33	1
18O2 PFHxS	116		50 - 150	04/30/20 11:33	04/30/20 16:33	1

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Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-22815-1

Client Sample ID: Effluent

Lab Sample ID: 190-22815-2

Date Collected: 04/22/20 14:45

Matrix: Water

Date Received: 04/23/20 10:24

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4:2 FTS	<16		16	ng/L		04/30/20 11:33	04/30/20 16:42	1
6:2 FTS	<16		16	ng/L		04/30/20 11:33	04/30/20 16:42	1
8:2 FTS	<16		16	ng/L		04/30/20 11:33	04/30/20 16:42	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<16		16	ng/L		04/30/20 11:33	04/30/20 16:42	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<16		16	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluorobutanesulfonic acid (PFBS)	3.1		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluorobutanoic acid (PFBA)	2.7		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluorodecanesulfonic acid (PFDS)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluorodecanoic acid (PFDA)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluorododecanoic acid (PFDoA)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluoroheptanesulfonic Acid (PFHpS)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluoroheptanoic acid (PFHpA)	1.7		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluorohexanesulfonic acid (PFHxS)	1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluorohexanoic acid (PFHxA)	13		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluorononanesulfonic acid (PFNS)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluorononanoic acid (PFNA)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluorooctanesulfonamide (FOSA)	<8.1		8.1	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluorooctane sulfonate (PFOS)	2.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluorooctanoic acid (PFOA)	4.4		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluoropentanesulfonic acid (PFPeS)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluoropentanoic acid (PFPeA)	4.2		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluorotetradecanoic acid (PFTeA)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluorotridecanoic acid (PFTriA)	<1.6 *		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1
Perfluoroundecanoic acid (PFUnA)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:42	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	78		25 - 150	04/30/20 11:33	04/30/20 16:42	1
13C4 PFBA	92		25 - 150	04/30/20 11:33	04/30/20 16:42	1
13C3 PFBS	98		50 - 150	04/30/20 11:33	04/30/20 16:42	1
13C2 PFDA	90		50 - 150	04/30/20 11:33	04/30/20 16:42	1
13C2 PFDoA	101		50 - 150	04/30/20 11:33	04/30/20 16:42	1
13C4 PFHpA	112		50 - 150	04/30/20 11:33	04/30/20 16:42	1
13C2 PFHxA	114		50 - 150	04/30/20 11:33	04/30/20 16:42	1
13C5 PFNA	99		50 - 150	04/30/20 11:33	04/30/20 16:42	1
13C4 PFOA	103		50 - 150	04/30/20 11:33	04/30/20 16:42	1
13C4 PFOS	101		50 - 150	04/30/20 11:33	04/30/20 16:42	1
13C5-PFPeA DNU	111		25 - 150	04/30/20 11:33	04/30/20 16:42	1
13C2 PFTeDA	58		50 - 150	04/30/20 11:33	04/30/20 16:42	1
13C2 PFUnA	89		50 - 150	04/30/20 11:33	04/30/20 16:42	1
d5-NEtFOSAA	93		50 - 150	04/30/20 11:33	04/30/20 16:42	1
d3-NMeFOSAA	89		50 - 150	04/30/20 11:33	04/30/20 16:42	1
M2-4:2 FTS	155	*5	25 - 150	04/30/20 11:33	04/30/20 16:42	1
M2-6:2 FTS	138		25 - 150	04/30/20 11:33	04/30/20 16:42	1
M2-8:2 FTS	116		25 - 150	04/30/20 11:33	04/30/20 16:42	1
18O2 PFHxS	108		50 - 150	04/30/20 11:33	04/30/20 16:42	1

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Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-22815-1

Client Sample ID: Effluent Duplic.

Lab Sample ID: 190-22815-3

Date Collected: 04/22/20 14:50

Matrix: Water

Date Received: 04/23/20 10:24

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4:2 FTS	<16		16	ng/L		04/30/20 11:33	04/30/20 16:50	1
6:2 FTS	<16		16	ng/L		04/30/20 11:33	04/30/20 16:50	1
8:2 FTS	<16		16	ng/L		04/30/20 11:33	04/30/20 16:50	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<16		16	ng/L		04/30/20 11:33	04/30/20 16:50	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<16		16	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluorobutanesulfonic acid (PFBS)	2.8		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluorobutanoic acid (PFBA)	2.5		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluorodecanesulfonic acid (PFDS)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluorodecanoic acid (PFDA)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluorododecanoic acid (PFDoA)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluoroheptanesulfonic Acid (PFHpS)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluoroheptanoic acid (PFHpA)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluorohexanesulfonic acid (PFHxS)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluorohexanoic acid (PFHxA)	13		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluorononanesulfonic acid (PFNS)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluorononanoic acid (PFNA)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluorooctanesulfonamide (FOSA)	<8.1		8.1	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluorooctane sulfonate (PFOS)	2.7		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluorooctanoic acid (PFOA)	4.7		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluoropentanesulfonic acid (PFPeS)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluoropentanoic acid (PFPeA)	4.1		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluorotetradecanoic acid (PFTeA)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluorotridecanoic acid (PFTriA)	<1.6 *		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1
Perfluoroundecanoic acid (PFUnA)	<1.6		1.6	ng/L		04/30/20 11:33	04/30/20 16:50	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	73		25 - 150	04/30/20 11:33	04/30/20 16:50	1
13C4 PFBA	88		25 - 150	04/30/20 11:33	04/30/20 16:50	1
13C3 PFBS	97		50 - 150	04/30/20 11:33	04/30/20 16:50	1
13C2 PFDA	90		50 - 150	04/30/20 11:33	04/30/20 16:50	1
13C2 PFDoA	91		50 - 150	04/30/20 11:33	04/30/20 16:50	1
13C4 PFHpA	115		50 - 150	04/30/20 11:33	04/30/20 16:50	1
13C2 PFHxA	113		50 - 150	04/30/20 11:33	04/30/20 16:50	1
13C5 PFNA	96		50 - 150	04/30/20 11:33	04/30/20 16:50	1
13C4 PFOA	100		50 - 150	04/30/20 11:33	04/30/20 16:50	1
13C4 PFOS	92		50 - 150	04/30/20 11:33	04/30/20 16:50	1
13C5-PFPeA DNU	103		25 - 150	04/30/20 11:33	04/30/20 16:50	1
13C2 PFTeDA	49	*5	50 - 150	04/30/20 11:33	04/30/20 16:50	1
13C2 PFUnA	83		50 - 150	04/30/20 11:33	04/30/20 16:50	1
d5-NEtFOSAA	84		50 - 150	04/30/20 11:33	04/30/20 16:50	1
d3-NMeFOSAA	75		50 - 150	04/30/20 11:33	04/30/20 16:50	1
M2-4:2 FTS	148		25 - 150	04/30/20 11:33	04/30/20 16:50	1
M2-6:2 FTS	131		25 - 150	04/30/20 11:33	04/30/20 16:50	1
M2-8:2 FTS	107		25 - 150	04/30/20 11:33	04/30/20 16:50	1
18O2 PFHxS	103		50 - 150	04/30/20 11:33	04/30/20 16:50	1

Eurofins TestAmerica, Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-22815-1

Method: 537 (modified) - Fluorinated Alkyl Substances

Lab Sample ID: MB 200-154582/1-A

Matrix: Water

Analysis Batch: 154592

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 154582

Analyte	MB Result	MB Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
4:2 FTS	<20		20	ng/L		04/30/20 11:33	04/30/20 16:09	1
6:2 FTS	<20		20	ng/L		04/30/20 11:33	04/30/20 16:09	1
8:2 FTS	<20		20	ng/L		04/30/20 11:33	04/30/20 16:09	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<20		20	ng/L		04/30/20 11:33	04/30/20 16:09	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<20		20	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluorobutanesulfonic acid (PFBS)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluorobutanoic acid (PFBA)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluorodecanesulfonic acid (PFDS)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluorodecanoic acid (PFDA)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluorododecanoic acid (PFDoA)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluoroheptanesulfonic Acid (PFHpS)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluoroheptanoic acid (PFHpA)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluorohexanesulfonic acid (PFHxS)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluorohexanoic acid (PFHxA)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluorononanesulfonic acid (PFNS)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluorononanoic acid (PFNA)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluorooctanesulfonamide (FOSA)	<10		10	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluorooctane sulfonate (PFOS)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluorooctanoic acid (PFOA)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluoropentanesulfonic acid (PFPeS)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluoropentanoic acid (PFPeA)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluorotetradecanoic acid (PFTeA)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluorotridecanoic acid (PFTriA)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1
Perfluoroundecanoic acid (PFUnA)	<2.0		2.0	ng/L		04/30/20 11:33	04/30/20 16:09	1

Isotope Dilution	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	68		25 - 150	04/30/20 11:33	04/30/20 16:09	1
13C4 PFBA	118		25 - 150	04/30/20 11:33	04/30/20 16:09	1
13C3 PFBS	119		50 - 150	04/30/20 11:33	04/30/20 16:09	1
13C2 PFDA	100		50 - 150	04/30/20 11:33	04/30/20 16:09	1
13C2 PFDoA	91		50 - 150	04/30/20 11:33	04/30/20 16:09	1
13C4 PFHpA	113		50 - 150	04/30/20 11:33	04/30/20 16:09	1
13C2 PFHxA	127		50 - 150	04/30/20 11:33	04/30/20 16:09	1
13C5 PFNA	108		50 - 150	04/30/20 11:33	04/30/20 16:09	1
13C4 PFOA	111		50 - 150	04/30/20 11:33	04/30/20 16:09	1
13C4 PFOS	106		50 - 150	04/30/20 11:33	04/30/20 16:09	1
13C5-PFPeA DNU	122		25 - 150	04/30/20 11:33	04/30/20 16:09	1
13C2 PFTeDA	80		50 - 150	04/30/20 11:33	04/30/20 16:09	1
13C2 PFUnA	89		50 - 150	04/30/20 11:33	04/30/20 16:09	1
d5-NEtFOSAA	87		50 - 150	04/30/20 11:33	04/30/20 16:09	1
d3-NMeFOSAA	90		50 - 150	04/30/20 11:33	04/30/20 16:09	1
M2-4:2 FTS	117		25 - 150	04/30/20 11:33	04/30/20 16:09	1
M2-6:2 FTS	106		25 - 150	04/30/20 11:33	04/30/20 16:09	1
M2-8:2 FTS	105		25 - 150	04/30/20 11:33	04/30/20 16:09	1
18O2 PFHxS	122		50 - 150	04/30/20 11:33	04/30/20 16:09	1

Eurofins TestAmerica, Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-22815-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 200-154582/2-A

Matrix: Water

Analysis Batch: 154592

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 154582

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
4:2 FTS	37.4	38.2		ng/L		102	50 - 150
6:2 FTS	37.9	33.2		ng/L		88	50 - 150
8:2 FTS	38.3	36.8		ng/L		96	50 - 150
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	40.0	36.7		ng/L		92	70 - 130
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	40.0	40.6		ng/L		102	70 - 130
Perfluorobutanesulfonic acid (PFBS)	35.4	35.2		ng/L		100	70 - 130
Perfluorobutanoic acid (PFBA)	40.0	39.0		ng/L		98	50 - 150
Perfluorodecanesulfonic acid (PFDS)	38.6	33.8		ng/L		88	50 - 150
Perfluorodecanoic acid (PFDA)	40.0	40.4		ng/L		101	70 - 130
Perfluorododecanoic acid (PFDoA)	40.0	37.6		ng/L		94	70 - 130
Perfluoroheptanesulfonic Acid (PFHpS)	38.1	36.9		ng/L		97	50 - 150
Perfluoroheptanoic acid (PFHpA)	40.0	37.5		ng/L		94	70 - 130
Perfluorohexanesulfonic acid (PFHxS)	36.4	33.3		ng/L		91	70 - 130
Perfluorohexanoic acid (PFHxA)	40.0	39.2		ng/L		98	70 - 130
Perfluorononanesulfonic acid (PFNS)	38.4	35.9		ng/L		93	50 - 150
Perfluorononanoic acid (PFNA)	40.0	41.6		ng/L		104	70 - 130
Perfluorooctanesulfonamide (FOSA)	40.0	34.4		ng/L		86	50 - 150
Perfluorooctane sulfonate (PFOS)	37.1	34.7		ng/L		94	70 - 130
Perfluorooctanoic acid (PFOA)	40.0	38.4		ng/L		96	70 - 130
Perfluoropentanesulfonic acid (PFPeS)	37.5	36.3		ng/L		97	50 - 150
Perfluoropentanoic acid (PFPeA)	40.0	36.6		ng/L		91	50 - 150
Perfluorotetradecanoic acid (PFTeA)	40.0	37.6		ng/L		94	70 - 130
Perfluorotridecanoic acid (PFTriA)	40.0	27.1 *		ng/L		68	70 - 130
Perfluoroundecanoic acid (PFUnA)	40.0	37.4		ng/L		94	70 - 130

Isotope Dilution	LCS %Recovery	LCS Qualifier	Limits
13C8 FOSA	69		25 - 150
13C4 PFBA	115		25 - 150
13C3 PFBS	110		50 - 150
13C2 PFDA	102		50 - 150
13C2 PFDoA	92		50 - 150
13C4 PFHpA	113		50 - 150
13C2 PFHxA	116		50 - 150
13C5 PFNA	104		50 - 150
13C4 PFOA	103		50 - 150
13C4 PFOS	107		50 - 150
13C5-PFPeA DNU	116		25 - 150
13C2 PFTeDA	79		50 - 150

Eurofins TestAmerica, Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-22815-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 200-154582/2-A

Matrix: Water

Analysis Batch: 154592

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 154582

Isotope Dilution	LCS	LCS	Limits
	%Recovery	Qualifier	
<i>13C2 PFUnA</i>	90		50 - 150
<i>d5-NEtFOSAA</i>	89		50 - 150
<i>d3-NMeFOSAA</i>	87		50 - 150
<i>M2-4:2 FTS</i>	110		25 - 150
<i>M2-6:2 FTS</i>	113		25 - 150
<i>M2-8:2 FTS</i>	97		25 - 150
<i>18O2 PFHxS</i>	116		50 - 150

Isotope Dilution Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-22815-1

Method: 537 (modified) - Fluorinated Alkyl Substances

Matrix: Water

Prep Type: Total/NA

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFOSA (25-150)	PFBA (25-150)	C3PFBS (50-150)	PFDA (50-150)	PFDoA (50-150)	C4PFHA (50-150)	PFHxA (50-150)	PFNA (50-150)
190-22815-1	Field Blank	67	113	110	93	83	111	117	99
190-22815-2	Effluent	78	92	98	90	101	112	114	99
190-22815-3	Effluent Duplic.	73	88	97	90	91	115	113	96
LCS 200-154582/2-A	Lab Control Sample	69	115	110	102	92	113	116	104
MB 200-154582/1-A	Method Blank	68	118	119	100	91	113	127	108

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFOA (50-150)	PFOS (50-150)	PFPeA (25-150)	PFTDA (50-150)	PFUnA (50-150)	d5NEFOS (50-150)	d3NMFOS (50-150)	M242FTS (25-150)
190-22815-1	Field Blank	106	108	116	71	80	77	77	125
190-22815-2	Effluent	103	101	111	58	89	93	89	155 *5
190-22815-3	Effluent Duplic.	100	92	103	49 *5	83	84	75	148
LCS 200-154582/2-A	Lab Control Sample	103	107	116	79	90	89	87	110
MB 200-154582/1-A	Method Blank	111	106	122	80	89	87	90	117

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	M262FTS (25-150)	M282FTS (25-150)	PFHxS (50-150)
190-22815-1	Field Blank	108	105	116
190-22815-2	Effluent	138	116	108
190-22815-3	Effluent Duplic.	131	107	103
LCS 200-154582/2-A	Lab Control Sample	113	97	116
MB 200-154582/1-A	Method Blank	106	105	122

Surrogate Legend

PFOSA = 13C8 FOSA
PFBA = 13C4 PFBA
C3PFBS = 13C3 PFBS
PFDA = 13C2 PFDA
PFDoA = 13C2 PFDoA
C4PFHA = 13C4 PFHpA
PFHxA = 13C2 PFHxA
PFNA = 13C5 PFNA
PFOA = 13C4 PFOA
PFOS = 13C4 PFOS
PFPeA = 13C5-PFPeA DNU
PFTDA = 13C2 PFTeDA
PFUnA = 13C2 PFUnA
d5NEFOS = d5-NEtFOSAA
d3NMFOS = d3-NMeFOSAA
M242FTS = M2-4:2 FTS
M262FTS = M2-6:2 FTS
M282FTS = M2-8:2 FTS
PFHxS = 18O2 PFHxS

Definitions/Glossary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-22815-1

Qualifiers

LCMS

Qualifier	Qualifier Description
*	LCS or LCSD is outside acceptance limits.
*5	Isotope dilution analyte is outside acceptance limits.

Glossary

Abbreviation	These commonly used abbreviations may or may not be present in this report.
□	Listed under the "D" column to designate that the result is reported on a dry weight basis
%R	Percent Recovery
CFL	Contains Free Liquid
CNF	Contains No Free Liquid
DER	Duplicate Error Ratio (normalized absolute difference)
Dil Fac	Dilution Factor
DL	Detection Limit (DoD/DOE)
DL, RA, RE, IN	Indicates a Dilution, Re-analysis, Re-extraction, or additional Initial metals/anion analysis of the sample
DLC	Decision Level Concentration (Radiochemistry)
EDL	Estimated Detection Limit (Dioxin)
LOD	Limit of Detection (DoD/DOE)
LOQ	Limit of Quantitation (DoD/DOE)
MDA	Minimum Detectable Activity (Radiochemistry)
MDC	Minimum Detectable Concentration (Radiochemistry)
MDL	Method Detection Limit
ML	Minimum Level (Dioxin)
NC	Not Calculated
ND	Not Detected at the reporting limit (or MDL or EDL if shown)
PQL	Practical Quantitation Limit
QC	Quality Control
RER	Relative Error Ratio (Radiochemistry)
RL	Reporting Limit or Requested Limit (Radiochemistry)
RPD	Relative Percent Difference, a measure of the relative difference between two points
TEF	Toxicity Equivalent Factor (Dioxin)
TEQ	Toxicity Equivalent Quotient (Dioxin)

QC Association Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-22815-1

LCMS

Prep Batch: 154582

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-22815-1	Field Blank	Total/NA	Water	3535	
190-22815-2	Effluent	Total/NA	Water	3535	
190-22815-3	Effluent Duplic.	Total/NA	Water	3535	
MB 200-154582/1-A	Method Blank	Total/NA	Water	3535	
LCS 200-154582/2-A	Lab Control Sample	Total/NA	Water	3535	

Analysis Batch: 154592

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-22815-1	Field Blank	Total/NA	Water	537 (modified)	154582
190-22815-2	Effluent	Total/NA	Water	537 (modified)	154582
190-22815-3	Effluent Duplic.	Total/NA	Water	537 (modified)	154582
MB 200-154582/1-A	Method Blank	Total/NA	Water	537 (modified)	154582
LCS 200-154582/2-A	Lab Control Sample	Total/NA	Water	537 (modified)	154582

Lab Chronicle

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-22815-1

Client Sample ID: Field Blank

Date Collected: 04/22/20 14:40

Date Received: 04/23/20 10:24

Lab Sample ID: 190-22815-1

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			154582	04/30/20 11:33	ND	TAL BUR
Total/NA	Analysis	537 (modified)		1	154592	04/30/20 16:33	BWC	TAL BUR

Client Sample ID: Effluent

Date Collected: 04/22/20 14:45

Date Received: 04/23/20 10:24

Lab Sample ID: 190-22815-2

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			154582	04/30/20 11:33	ND	TAL BUR
Total/NA	Analysis	537 (modified)		1	154592	04/30/20 16:42	BWC	TAL BUR

Client Sample ID: Effluent Duplic.

Date Collected: 04/22/20 14:50

Date Received: 04/23/20 10:24

Lab Sample ID: 190-22815-3

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			154582	04/30/20 11:33	ND	TAL BUR
Total/NA	Analysis	537 (modified)		1	154592	04/30/20 16:50	BWC	TAL BUR

Laboratory References:

TAL BUR = Eurofins TestAmerica, Burlington, 30 Community Drive, Suite 11, South Burlington, VT 05403, TEL (802)660-1990

Analyst References:

Lab: TAL BUR

Batch Type: Prep

ND = Nathan Dean

Batch Type: Analysis

BWC = Bradley Chirgwin

Accreditation/Certification Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-22815-1

Laboratory: Eurofins TestAmerica, Michigan

All accreditations/certifications held by this laboratory are listed. Not all accreditations/certifications are applicable to this report.

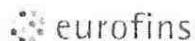
Authority	Program	Identification Number	Expiration Date
Michigan	State	0057	10-01-20

Laboratory: Eurofins TestAmerica, Burlington

All accreditations/certifications held by this laboratory are listed. Not all accreditations/certifications are applicable to this report.

Authority	Program	Identification Number	Expiration Date
ANAB	Dept. of Defense ELAP	L2336	02-25-23
Connecticut	State	PH-0751	09-30-21
DE Haz. Subst. Cleanup Act (HSCA)	State	N/A	05-15-20
Florida	NELAP	E87467	06-30-20
Minnesota	NELAP	050-999-436	12-31-20
New Hampshire	NELAP	2006	12-18-20
New Jersey	NELAP	VT972	06-30-20
New York	NELAP	10391	04-01-21
Pennsylvania	NELAP	68-00489	04-30-21
Rhode Island	State	LAO00298	12-30-20
US Fish & Wildlife	US Federal Programs	058448	07-31-20
USDA	US Federal Programs	P330-17-00272	08-09-20
Vermont	State	VT4000	12-31-20
Virginia	NELAP	460209	12-14-20

Ver: 01/16/2019



Environment Testing
TestAmerica

☐ SDS or Known Hazard Information Supplied by Client

☐ Discrepancies

Client ID: City of Cadillac

☐ Short Hold

Work Order #: 190-22815

☐ Rush ☐ 24 Hr ☐ 2-Day ☐ 3-Day ☐ 5-Day ☐ Other:

Receipt Evaluation Performed by: Initials: TEH Date: 4/23/20 Time: 1024

Cooler / Sample Receipt

After hours receipt: complete gray

areas. Place cooler in walk-in, place

form in Receiving box. Date: _____ Time: _____

Method of Shipment:

Walk-In Client Eurofins TA Field/Courier

Other Client / 3rd Party Courier: _____

Fed Ex Tracking #: _____

UPS Tracking #: 1Z E47 441 03 5479 5501

Other: _____

Shipping Container Type:

☒ Cooler ☐ Box

☐ None ☐ Other: _____

Packing Materials:

☒ Plastic Bags ☐ Foam

☐ Bubble Wrap ☐ Paper

☐ Packing Peanuts ☐ None

☐ Other: _____

Custody Seals Intact:

☒ Yes ☐ No

☐ NA (not used or required)

Cooling Materials:

☐ Ice (Solid) ☒ Ice (Melted)

☐ Blue Ice ☐ None

☐ Other: _____

Bacteriological Samples	Temp Corrected (°C)	Frozen?		Rec'd Within 2 Hrs?		Sample Flagged?	
		Yes	No	Yes	No	Yes	No

Received on same day sampled? Yes No

Additional Sheets Required? Yes No

Receipt Temperatures

Thermometer ID	Observed (°C)	Corrected (°C)	Temp Blank	Sample Temp	Acceptable	Cooler ID	Affected Samples
<u>CP 313207</u>	<u>2.0</u>	<u>2.0</u>		☒	☒ Y ☐ N		
					☐ Y ☐ N		
					☐ Y ☐ N		

Receipt Questions**	Y	N	NA	"No" answers require additional comment
CoC present and ETA receipt signature, date, and time properly documented?	☒			
Containers and Labels in good condition? (unbroken, not leaking, appropriately filled, labels legible & attached)	☒			
Appropriate containers used and adequate volume provided?	☒			Preserved bottles checked for pH?* Yes No
Number of sample containers match CoC?	☒			pH strip lot # _____
Samples received within hold?	☒			
Samples submitted for GRO and Volatiles analysis (8260, 624, 524) received without headspace?			☒	
Was a Trip Blank received with VOA samples?			☒	
Were the samples free of any questionable physical conformities? (i.e.; field duplicates or multiple bottles of the same sample do not significantly vary in appearance - color, solid proportions, etc.)	☒			
Were the CoC bottle labels and all other items free of all other discrepancies or issues that would need to be addressed with the Project Manager and/or Client?	☒			
**May not be applicable if samples are not for compliance testing				*Excludes FOG, VOAs, TOC Vials, HEM

Client Contact Record

Contact Via: ☐ Phone ☐ Email ☐ Other: _____ Person Contacted: _____ Date/Time: _____

☐ Discrepancy allowance agreement is on record in the client project file

Discussion / Resolution

Any additional documentation and clarification from the client must be noted in the narrative and/or scanned into the CoC directory.

Reviewed by Jim Hall Date: 4/23/20

WI-MI-010_020720

Environment Testing
TestAmerica

Ver: 01/16/2019

[illegible]

ORIGIN ID:DEOA (810) 229-2763
SHIPPING DEPARTMENT
EUROFINS TESTAMERICA
10448 CITATION DRIVE
SUITE 200
BRIGHTON, MI 48116
UNITED STATES US

SHIP DATE: 23APR20
ACTWGT: 34.80 LB
CAD: 0183192/CAFE3313

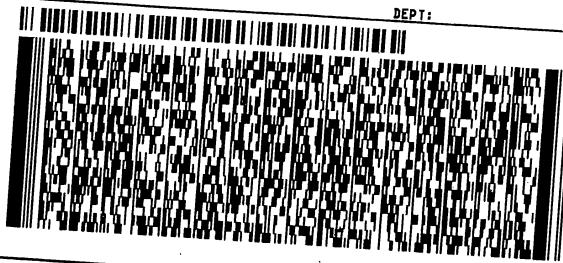
BILL RECIPIENT

TO **SAMPLE RECIEVING**
EUROFINS TEST AMERICA BURLINGTON
30 COMMUNITY DR
STE 11
SOUTH BURLINGTON VT 054036834

(802) 660-1990
INV:
PO:

REF:

DEPT:



FedEx
Express



JT91219082001W

TRK#
0201 4506 9514 1016

FRI - 24 APR 10:30A
PRIORITY OVERNIGHT

NL BTVA

05403
VT-US BTV

